

Module 9 — Rapid Recap

Exploratory Data Analysis (EDA)

One combined sentence per subtopic: meaning + difference + interview-ready phrasing

Topics 1–7, all 35 subtopics

Topic 1: Understand EDA

EDA means exploring your data carefully to understand it before doing any analysis or modeling.

1a. Purpose of EDA: EDA is the process of understanding data before using it - unlike cleaning, which fixes data, EDA's job is only to understand it - as you'd say: 'EDA is the process of understanding data before using it, so decisions are based on facts, not guesses.'

1b. Understanding datasets: Understanding a dataset means checking rows, columns, and data types first - a basic manual look, unlike profiling (2e) which automates this - as you'd say: 'Before analyzing anything, I check the dataset's shape, columns, and data types to know what I'm working with.'

1c. Identifying patterns: Identifying patterns means finding repeating trends that show what's 'normal' - the opposite of anomalies, which break that pattern - as you'd say: 'I look for patterns and correlations to understand relationships between features.'

1d. Detecting anomalies: Detecting anomalies means finding unusual/wrong values using IQR or Z-score - different from a rare-but-valid value, since anomalies are usually errors - as you'd say: 'I use IQR or Z-score to detect outliers or errors before modeling.'

1e. Data storytelling: Data storytelling means turning findings into a clear explanation stakeholders can act on - different from a chart alone, which only shows data - as you'd say: 'Data storytelling means translating numbers into a clear insight stakeholders can act on.'

Topic 2: Understand Descriptive Analysis

Descriptive analysis means summarizing data using numbers like average, spread, and shape to describe it briefly.

2a. Summary statistics: Summary statistics are a quick set of numbers describing a column at a glance - broader than central tendency, which is just the 'center' part - as you'd say: 'Summary statistics give a quick numerical overview using mean, min, max, and standard deviation.'

2b. Distribution analysis: Distribution analysis looks at the shape of spread (symmetric, skewed) - different from variability, which measures how much spread, not shape - as you'd say: 'Distribution analysis helps me see whether data is symmetric or skewed, affecting preprocessing choices.'

2c. Variability analysis: Variability analysis measures consistency using one number like std dev - different from distribution, which shows the whole shape - as you'd say: 'Variability analysis, using std dev or IQR, tells me how consistent or scattered the data is.'

2d. Central tendency analysis: Central tendency finds the single typical value (mean/median/mode) - narrower than summary stats, which also includes spread - as you'd say: 'I use mean, median, or mode depending on skew to represent the typical value.'

2e. Dataset profiling: Dataset profiling bundles every check into one report - different from 1b's manual look, since profiling is automated - as you'd say: 'Dataset profiling combines all checks into one comprehensive report before analysis.'

Topic 3: Understand Univariate Analysis

Univariate analysis means studying just one variable at a time, on its own, without comparing it to any other variable.

3a. Single-variable analysis: Studying one column in isolation is the foundational first step - different from bivariate/multivariate since no other variable is involved - as you'd say: 'I always analyze each variable individually before checking relationships between them.'

3b. Histograms: A histogram visually shows value distribution shape - different from a frequency table since it's graphical, not tabular - as you'd say: 'I use histograms to visually inspect the shape and skew of a numeric variable.'

3c. Frequency distributions: A frequency distribution counts occurrences per value/category - applies to both numeric and categorical data as counts, not shape - as you'd say: 'I check frequency distributions to understand class balance in categorical features.'

3d. Outlier detection: Outlier detection finds extreme values in one column using box plots or IQR - same concept as anomaly detection (1d) applied within univariate analysis - as you'd say: 'I use IQR and box plots to detect outliers within individual features.'

3e. Distribution interpretation: Interpretation converts histogram shape and stats into an actionable conclusion - drives next steps like transforms - as you'd say: 'After analyzing a distribution, I interpret it to

decide if transformation or scaling is needed.'

Topic 4: Understand Bivariate Analysis

Bivariate analysis means studying the relationship between exactly two variables at a time.

4a. Feature relationships: Feature relationships means examining how two variables connect - the shift from univariate's single-variable focus - as you'd say: 'I check relationships between features to understand how variables might influence each other.'

4b. Correlation analysis: Correlation gives one number (-1 to +1) for linear relationship strength - narrower than feature relationships, which also covers categorical pairs - as you'd say: 'I use Pearson correlation to quantify strength and direction of linear relationships.'

4c. Scatter plots: A scatter plot visually shows the actual relationship shape - different from correlation, which gives only a number - as you'd say: 'I use scatter plots to visually confirm what a correlation coefficient suggests.'

4d. Comparative analysis: Comparative analysis compares a numeric variable across categorical groups - different from correlation/scatter which need two numeric variables - as you'd say: 'I use group-by comparisons and box plots to see how a numeric feature varies across categories.'

4e. Trend identification: Trend identification tracks directional patterns over an ordered sequence like time - different from a plain scatter, since it's sequence-based - as you'd say: 'I check for trends over time to see if a relationship is strengthening, weakening, or stable.'

Topic 5: Understand Multivariate Analysis

Multivariate analysis means studying the relationships between three or more variables at the same time.

5a. Multiple feature relationships: Studying how 3+ variables connect together rather than just pairs - different from bivariate's two-variable focus - as you'd say: 'I analyze combinations of features together since outcomes are rarely explained by one relationship.'

5b. Feature interaction analysis: Checks if one variable's effect changes depending on another - different from 5a since it's about effects influencing each other - as you'd say: 'I check for interactions to see if a feature's impact differs across subgroups.'

5c. Pattern discovery: Finds hidden groupings across multiple variables combined - different from 1c's simpler pattern-spotting - as you'd say: 'I look for multivariate patterns since some risk segments only appear when conditions combine.'

5d. Dimensional thinking: The mindset of treating each variable as an added dimension - conceptual, not a specific technique - as you'd say: 'I think of records as points in a multi-dimensional feature space, how models actually process data.'

5e. Business insights generation: Translates multivariate findings into an actionable recommendation - mirrors 1e/3e but for combined findings - as you'd say: 'I translate multivariate patterns into specific, actionable business recommendations.'

Topic 6: Understand Visualization Principles

Visualization principles are the rules and best practices for choosing and designing charts so they communicate data honestly and clearly.

6a. Choosing the right chart: Matching chart type to data and question - about selection, not design or accuracy - as you'd say: 'I pick chart types based on variable count and type - histograms, scatter plots, or bar charts.'

6b. Data storytelling (visual): Designing a chart to make its insight obvious via titles and highlights - about framing an already-chosen chart - as you'd say: 'I design charts with descriptive titles and highlighted key points so the insight is clear.'

6c. Visualization best practices: General hygiene rules - clear labels, scale, minimal clutter - apply to any chart regardless of narrative - as you'd say: 'I follow best practices like clear labels and reduced clutter to keep charts trustworthy.'

6d. Avoiding misleading visuals: Catching distortions like truncated axes - focuses on accuracy/honesty specifically - as you'd say: 'I check for distortions like non-zero axes to make sure a chart doesn't exaggerate differences.'

6e. Effective communication: The overarching goal of tailoring visuals to the audience - the audience-awareness layer over all other subtopics - as you'd say: 'I tailor the level of detail based on audience - technical teams get detail, executives get one takeaway.'

Topic 7: Understand ML-Oriented EDA

ML-Oriented EDA means doing exploratory analysis specifically to prepare data for machine learning.

7a. Target variable analysis: Studying the predicted column's type and balance - different from feature importance since it focuses only on the outcome - as you'd say: 'I check the target's balance and type first, since it determines model type and metric.'

7b. Feature importance awareness: Early EDA-level estimate of predictive features using correlation/group comparisons - different from formal importance, which needs a trained model - as you'd say: 'I use correlation and group differences to get an early sense of predictive features.'

7c. Bias identification: Checks for representation gaps or historical unfairness in the data - about fairness, not correctness - as you'd say: 'I check for representation gaps and historical bias since a model can be accurate but unfair.'

7d. Data quality validation: Confirms the dataset is technically ready - no missing values, duplicates, or leakage - about correctness/usability, not fairness - as you'd say: 'I validate for missing values, duplicates, and

leakage since these directly break model performance.'

7e. Preparing for model building: Consolidates every finding into a final readiness checklist - the capstone step of Topic 7 and Module 9 - as you'd say: 'I summarize target balance, feature signals, bias checks, and data quality into one readiness checklist.'
